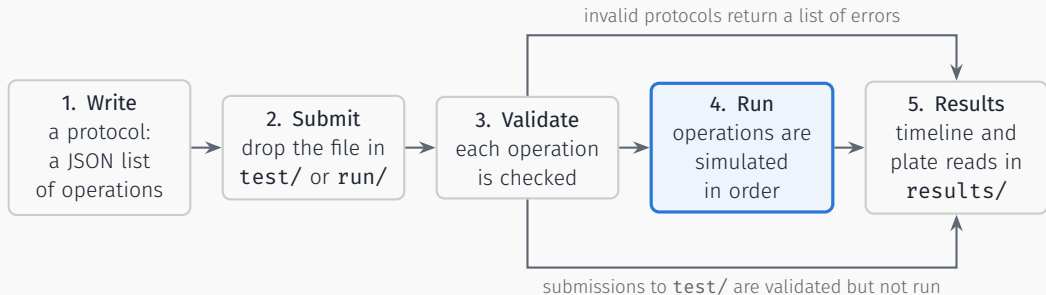


CHESSSim Tutorial

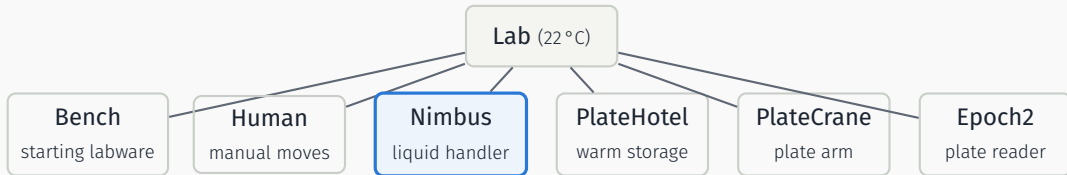
Autonomous Science

Workflow



- The organism is *Streptococcus gulonis* A2. Its growth requirements are unknown.
- The only data returned are plate reads from the Epoch2.
- Each submission starts with a new lab in the same initial state.

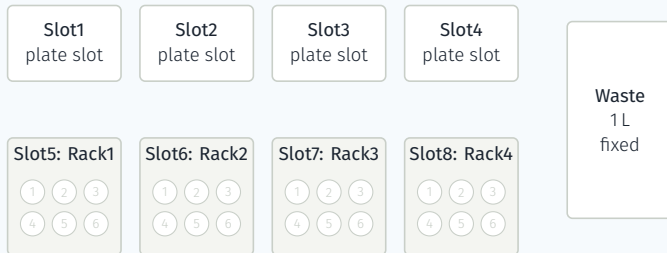
Lab layout



Locations are nested. The **path** to a location lists each parent, separated by /, e.g.
`Nimbus/Slot5/Rack1/Slot2`

Nimbus

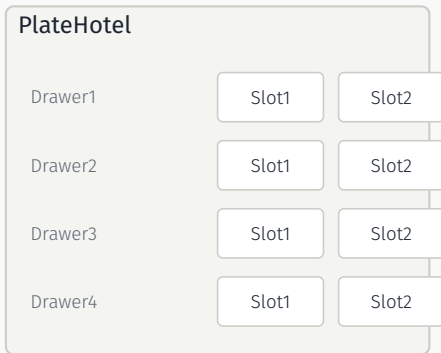
Nimbus



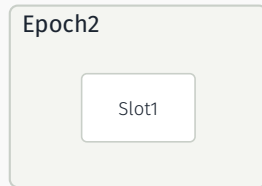
each rack has tube slots Slot1–Slot6; racks and Waste cannot be moved

The Nimbus has one **SingleChannel** head that transfers **2–1000 μ L** between any wells on the deck.
The waste bottle is **Nimbus/Waste/A1**.

PlateHotel and Epoch2



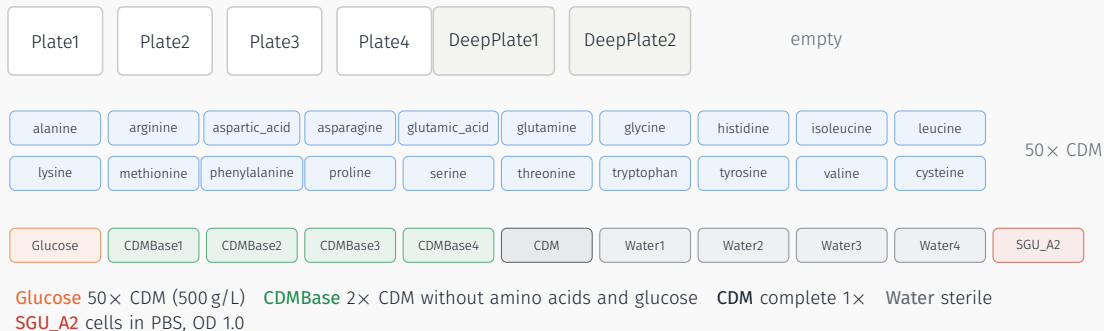
Incubator. The temperature is set with **set_attribute**. Plates are loaded by the PlateCrane.



Absorbance plate reader. A **read** measures all 96 wells of the plate in **Epoch2/Slot1**.

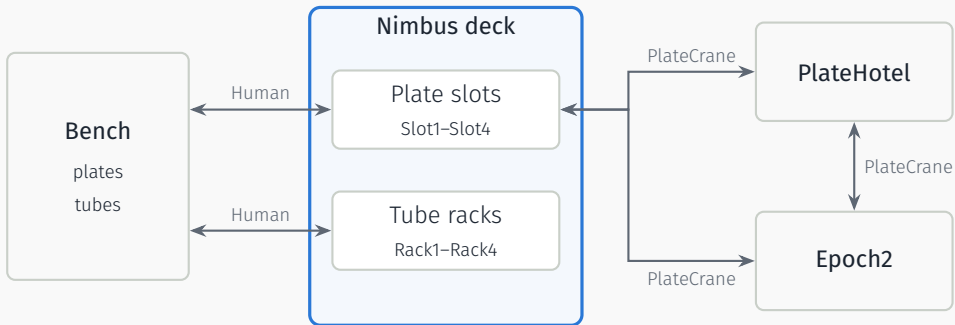
Both hold standard 96-well plates only (no deep-well plates).

Bench



The Bench has 31 tubes with 45 mL each. The Nimbus racks hold 24 tubes.

Movements



Plates move from the Bench to the Nimbus, and then to the PlateHotel or Epoch2. Tubes can only be moved to the Nimbus racks.

Instruments

Instrument	Operation	Head	Type	Reaches
Nimbus	transfer	SingleChannel	liquid handler	its own deck: plates, tubes, Waste
Human	move	Hands	a person	Bench and Nimbus deck
PlateCrane	move	Gripper	plate arm	Nimbus Slot1–Slot4, PlateHotel, Epoch2
PlateHotel	set_attribute	none	warm storage	itself (Temperature only)
Epoch2	read	none	plate reader	the plate in its slot (Absorbance)

All operations specify an **instrument**. Moves and transfers also specify a **head**.

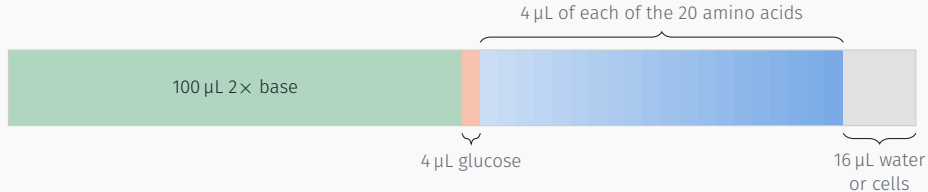
Labware and media

Labware

Labware	Names	Wells	Capacity	Locations
96-well plate	Plate1–Plate4	A1–H12	400 μ L	Bench, plate slots, hotel, reader
Deep-well plate	DeepPlate1–DeepPlate2	A1–H12	2 mL	Bench, plate slots
50 mL tube	31 named tubes	A1	50 mL	Bench, rack slots
Waste bottle	Waste	A1	1 L	fixed on the Nimbus

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B												
C												
D												
E												
F												
G												
H												

A1: row letter, column number



1 $\times$ CDM (stocks are 50 $\times$)

- glucose 10 g/L
- cysteine 0.65 g/L, threonine 0.2 g/L
- the other 18 amino acids 0.1 g/L each

CDM base (stocks are 2 $\times$) phosphates, acetate, bicarbonate, MgSO_4 , MnSO_4 , CaCl_2 , iron, adenine, guanine, uracil, and vitamins (riboflavin, NADPH, pantothenate, niacinamide, pyridoxal, pyridoxamine, thiamine, folate, biotin, PABA, B12). Full recipe in the lab manual.

Rules

Placement

	Bench	Nimbus Slot1-4	Rack slots	PlateHotel	Epoch2
96-well plate	✓	✓	✗	✓	✓
Deep-well plate	✓	✓	✗	✗	✗
50 mL tube	✓	✗	✓	✗	✗

- Each slot holds one item. Racks and Waste cannot be moved.
- Only plates and tubes can be moved.
- Both locations in a move must be reachable by the instrument.

Transfers, reads, and timing

Transfers

- 2–1000 μL per transfer
- maximum volumes: 400 μL (well), 2 mL (deep well), 50 mL (tube), 1 L (waste)
- the donor must contain enough liquid
- transfers take a proportional amount of every component, including cells

Reads and temperature

- plates are read in **Epoch2/Slot1**
- only the PlateHotel temperature can be set
- wells are at the temperature of their current location

Timing

- operations run sequentially

op 1: 170 μL $\rightarrow$ A1



op 2: 170 μL $\rightarrow$ A1



op 3: 170 μL $\rightarrow$ A1



volumes are checked in order

Protocols

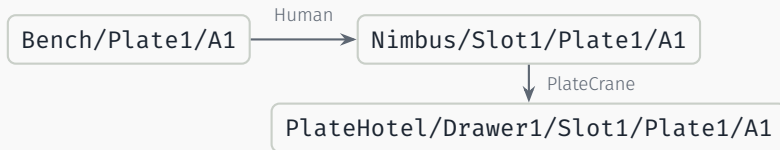
Protocol format

```
{ "operations": [  
  { "op": "move", "child": "Bench/Plate1", "parent": "Nimbus/Slot1", "instrument": "Human", "  
    head": "Hands" },  
  { "op": "pause", "duration": 10, "unit": "minute" }  
]}
```

- Operations are run in the order listed.
- Protocols can be written by hand or generated with any language (recommended).
- Operations can include an optional **"comment"** field.
- Errors and results refer to operations by their index, starting at 1.

Nimbus / Slot5 / Rack1 / Slot2 / SGU_A2 / A1
instrument deck slot rack tube slot tube well

Paths change when labware is moved



Each operation must use the current path of the labware.

Move and transfer

move a plate or tube (**child**) to a new location (**parent**)

```
{ "op": "move", "child": "Bench/SGU_A2", "parent": "Nimbus/Slot5/Rack1/Slot2", "instrument":  
  "Human", "head": "Hands" }  
{ "op": "move", "child": "Nimbus/Slot1/Plate1", "parent": "Epoch2/Slot1", "instrument": "  
  PlateCrane", "head": "Gripper" }
```

transfer liquid from a **donor** well to a **recipient** well (both on the Nimbus deck)

```
{ "op": "transfer", "donor": "Nimbus/Slot5/Rack1/Slot1/CDM/A1", "recipient": "Nimbus/Slot1/  
  Plate1/A1",  
  "quantity": 196, "unit": "µL", "instrument": "Nimbus", "head": "SingleChannel" }
```

Set attribute, read, and pause

set_attribute sets the PlateHotel temperature

```
{"op": "set_attribute", "location": "PlateHotel", "attribute": "Temperature", "value": 37, "unit": "°C", "instrument": "PlateHotel"}
```

read measures the absorbance of every well (**kind** is optional)

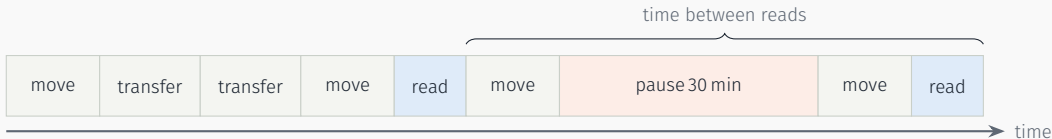
```
{"op": "read", "location": "Epoch2/Slot1/Plate1", "kind": "Absorbance", "instrument": "Epoch2"}
```

pause waits for a fixed time

```
{"op": "pause", "duration": 30, "unit": "minute"}
```

Units: μL (or uL), mL ; $^{\circ}\text{C}$ (or C); s, min, h.

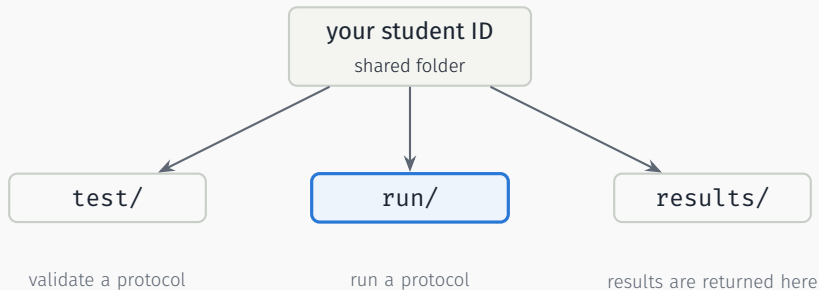
Timing



- Protocols do not specify times. Each operation's duration is set by the instrument, and the next operation starts when it finishes.
- Use **pause** to wait, e.g. between reads.
- The results include a **timeline** with the start and finish of each operation. Submit to **test/** to get the timeline without running the protocol.

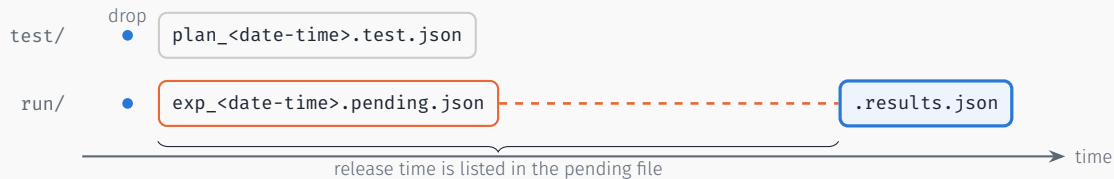
Submission

Submission folders



Save the protocol as a `.json` file and copy it to `test/` or `run/`. The file is removed from the folder when the server receives it.

Tests and runs



test/

- validation and timeline only
- no reads
- results returned immediately

run/

- full simulation with reads
- results returned after a delay
- invalid protocols are returned immediately

Run queue

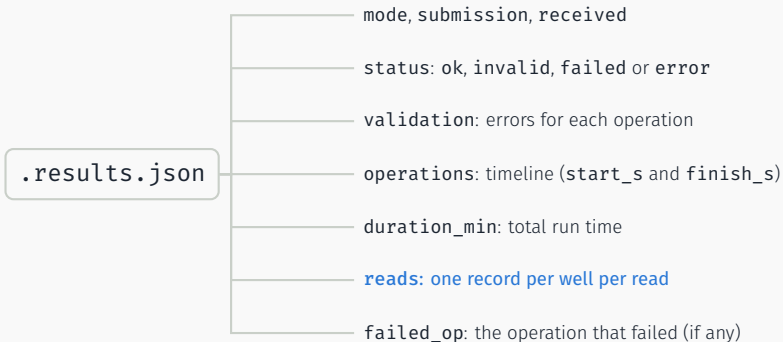


Each student can have one run in progress. Additional runs are queued, and the pending file lists the expected release time.

```
{"status": "pending", "mode": "run", "submission": "exp.json",  
  "received": "2026-09-30T22:47:15", "release_at": "2026-09-30T23:21:03"}
```

Results

Results files



Test results (**.test.json**) have the same format without **reads**. Files that cannot be parsed return **status error** with a **message**.

Reads

```
{
  "op_index": 11,
  "time_min": 10.633333333333333,
  "sim_time": "2026-01-05T09:10:38",
  "instrument": "Epoch2",
  "location": "Epoch2/Slot1/Plate1",
  "well": "Epoch2/Slot1/Plate1/A1",
  "kind": "Absorbance",
  "value": 0.060004737151923004,
  "unit": "OD",
  "environment": {"Temperature":
    {"value": 22.0, "unit": "°C"}}
}
```

- Each read returns 96 records, one per well.
- **time_min** is the time since the start of the run.
- **environment** is the well's environment during the read. Plates in the Epoch2 are at room temperature.
- The well path changes as the plate moves. Use the well name (**A1**) to match reads over time.

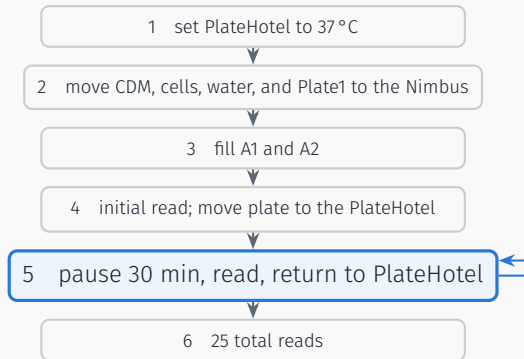
Example 1: Growth curve

Experimental design

	1	2	3	4	5	6	7	8	9	10	11	12
A	●	○	○	○	○	○	○	○	○	○	○	○
B	○	○	○	○	○	○	○	○	○	○	○	○
C	○	○	○	○	○	○	○	○	○	○	○	○
D	○	○	○	○	○	○	○	○	○	○	○	○
E	○	○	○	○	○	○	○	○	○	○	○	○
F	○	○	○	○	○	○	○	○	○	○	○	○
G	○	○	○	○	○	○	○	○	○	○	○	○
H	○	○	○	○	○	○	○	○	○	○	○	○

A1 196 μ L CDM + 4 μ L cells (OD 0.02)

A2 196 μ L CDM + 4 μ L water (control)



Protocol: setup

```
{ "op": "set_attribute", "location": "PlateHotel", "attribute": "Temperature", "value": 37, "unit": "°C",  
  "instrument": "PlateHotel" },  
{ "op": "move", "child": "Bench/CDM", "parent": "Nimbus/Slot5/Rack1/Slot1", HUMAN },  
{ "op": "move", "child": "Bench/SGU_A2", "parent": "Nimbus/Slot5/Rack1/Slot2", HUMAN },  
{ "op": "move", "child": "Bench/Water1", "parent": "Nimbus/Slot5/Rack1/Slot3", HUMAN },  
{ "op": "move", "child": "Bench/Plate1", "parent": "Nimbus/Slot1", HUMAN },  
{ "op": "transfer", "donor": "Nimbus/Slot5/Rack1/Slot1/CDM/A1", "recipient": "Nimbus/Slot1/Plate1/A1",  
  "quantity": 196, "unit": "µL", NIMBUS },  
{ "op": "transfer", "donor": "Nimbus/Slot5/Rack1/Slot2/SGU_A2/A1", "recipient": "Nimbus/Slot1/Plate1/A1",  
  "quantity": 4, "unit": "µL", NIMBUS },  
{ "op": "transfer", "donor": "Nimbus/Slot5/Rack1/Slot1/CDM/A1", "recipient": "Nimbus/Slot1/Plate1/A2",  
  "quantity": 196, "unit": "µL", NIMBUS },  
{ "op": "transfer", "donor": "Nimbus/Slot5/Rack1/Slot3/Water1/A1", "recipient": "Nimbus/Slot1/Plate1/A2",  
  "quantity": 4, "unit": "µL", NIMBUS },  
{ "op": "move", "child": "Nimbus/Slot1/Plate1", "parent": "Epoch2/Slot1", CRANE },  
{ "op": "read", "location": "Epoch2/Slot1/Plate1", "kind": "Absorbance", "instrument": "Epoch2" },  
{ "op": "move", "child": "Epoch2/Slot1/Plate1", "parent": "PlateHotel/Drawer1/Slot1", CRANE }
```

HUMAN "instrument": "Human", "head": "Hands"

NIMBUS "instrument": "Nimbus", "head": "SingleChannel"

CRANE "instrument": "PlateCrane", "head": "Gripper"

Protocol: repeated reads

```
hotel, reader = "PlateHotel/Drawer1/Slot1", "Epoch2/Slot1"

def move(child, parent):
    return {"op": "move", "child": child, "parent": parent,
            "instrument": "PlateCrane", "head": "Gripper"}

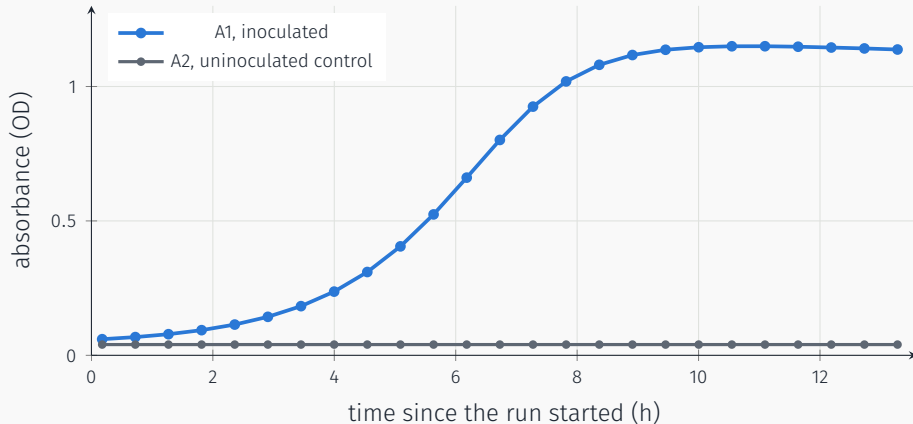
read = {"op": "read", "location": f"{reader}/Plate1", "kind": "Absorbance",
        "instrument": "Epoch2"}

for _ in range(24):
    ops += [{"op": "pause", "duration": 30, "unit": "minute"},
            move(f"{hotel}/Plate1", reader), read,
            move(f"{reader}/Plate1", hotel)]

json.dump({"operations": ops}, open("growth_curve.json", "w"), indent=2)
```

The complete script (108 operations) is in the lab manual.

Results

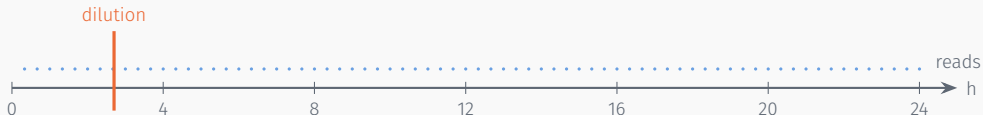
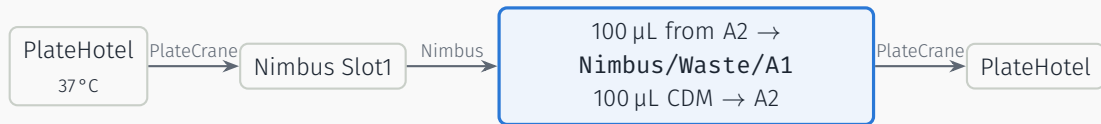


A1 grows from OD 0.06 to a maximum of 1.15. The control (A2) stays at 0.04, the same as an empty well.

Example 2: Dilution

Experimental design

A1 and A2 are inoculated in 200 μL CDM; A3 is an uninoculated control. The plate is read every 20 minutes for 24 hours. After 2.5 hours, half of A2 is replaced with fresh CDM.



Timing operations

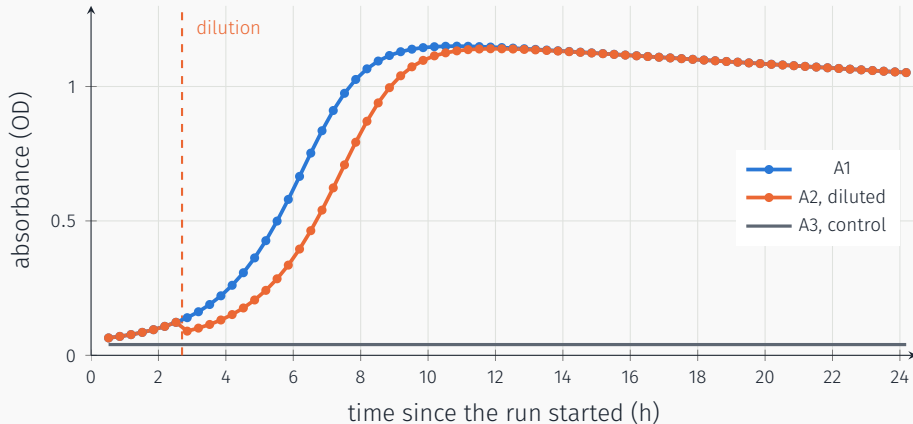
Operations cannot be scheduled at specific times. To read every 20 minutes or dilute at 2.5 hours, the protocol generator must track the time. Operation durations can be found from the timeline of a `test/` submission.

```
test = json.load(open("results/plan_20260930-101500.test.json"))
took = {}          # duration (s) of each operation type
for o in test["operations"]:
    r = o["requested"]
    took[(r["op"], r.get("instrument"))] = o["finish_s"] - o["start_s"]

clock = 0.0        # current time (s); add took[...] for each operation
def pause_until(t):
    global clock
    ops.append({"op": "pause", "duration": (t - clock) / 60, "unit": "minute"})
    clock = t
```

To read at time t , pause until t minus the duration of the move to the reader.

Results

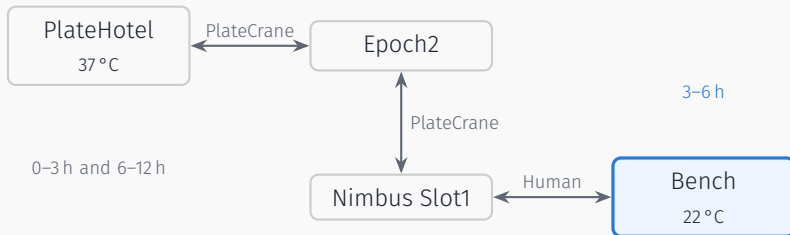


The dilution reduced A2 from OD 0.123 to 0.090. A2 then lags A1 by about an hour and reaches a maximum of 1.14.

Example 3: Plate left on the
bench

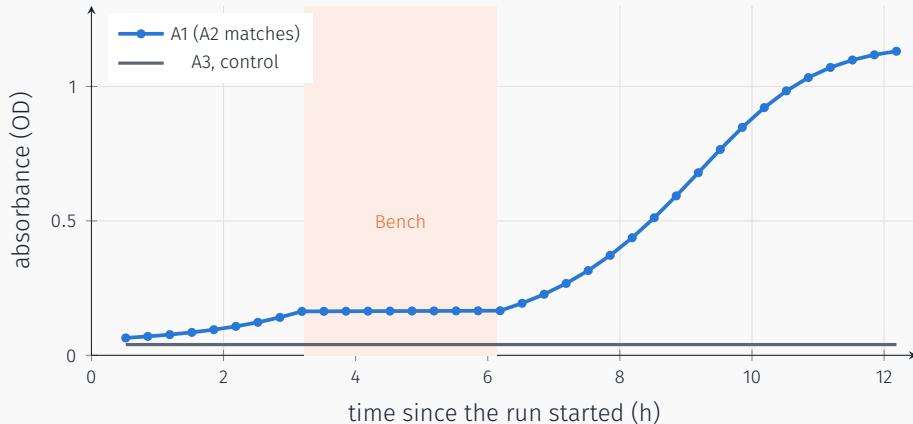
Experimental design

The plate is set up as in Example 2 and read every 20 minutes for 12 hours. From 3 to 6 hours, the plate is kept on the Bench at room temperature instead of the PlateHotel.



Moving the plate between the Bench and the Epoch2 requires both the Human and the PlateCrane.

Results



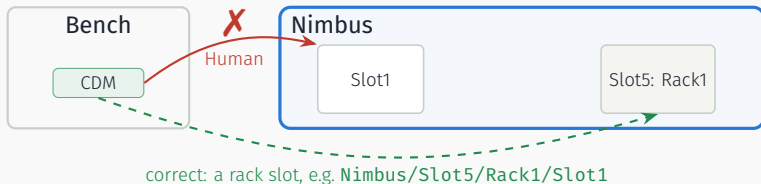
A1 does not grow on the Bench (OD 0.163 to 0.166 over 3 hours). Growth resumes after the plate returns to 37 °C.

Troubleshooting

Example: invalid move

A protocol that moves a 50 mL tube from the Bench to a Nimbus plate slot:

```
{ "operations": [  
  { "op": "move", "child": "Bench/CDM", "parent": "Nimbus/Slot1",  
    "instrument": "Human", "head": "Hands" }  
]}
```



Example: error report

Result of submitting the protocol to `test/`:
`results/tube_20261001-140505.test.json`

```
{
  "duration_min": 0.0,
  "epoch": "2026-01-05T09:00:00",
  "lab": "Lab",
  "mode": "test",
  "operations": [],
  "received": "2026-10-01T14:05:05",
  "status": "invalid",
  "submission": "tube.json",
  "validation": [
    {
      "index": 1,
      "message": "Occupancy Error: (2//1)\nmovement would overoccupy parent",
      "rule": "chess_error"
    }
  ]
}
```

- **status** is **invalid**, so nothing was run and **operations** is empty.
- Each entry in **validation** gives the operation **index** (here the first operation), the **rule**, and a **message**.
- **chess_error** means CHESS rejected the operation. The message says that the tube does not fit in the slot: plate slots hold plates only.
- The same protocol submitted to **run/** returns the same report immediately.

Validation errors

Error	Cause	Fix
unknown_location	the path does not exist	check spelling; use the current path after a move
unreachable volume_out_of_range	the instrument cannot reach a location volume is below 2 μ L or above 1000 μ L	see the Movements slide split large transfers; dilute stocks for small ones
wrong_head_type	wrong head for the operation	transfers use the Nimbus; moves use the Human or PlateCrane
not_in_instrument	location is not in the instrument	read in Epoch2/Slot1 ; set the PlateHotel
empty_donor chess_error	the donor well is empty slot occupied, labware does not fit, or well overflow	fill the donor first see the message and the Placement slide

If a run fails, the results include all reads before the failure and the index of the failed operation (**failed_op**).

Checklist

1. Generate the protocol with a script.
2. Keep track of each plate's location.
3. Include uninoculated controls.
4. Submit to **test/**.
5. Check the timing of the reads in the timeline.
6. Submit to **run/**.

